

Quiz 6

ECE 2201 – Spring 2025

Instructions

- Do not open the quiz until the instructor announces the start
- You will have 15 minutes to complete the quiz
- Closed book and notes – no formula sheets
- You may use your calculator
- Annotate your units with square brackets → for example, [A] for Amps
- Write your answer in the space provided

Name: Solution PS ID: _____

Determine the output voltage for the circuit and specify the linear range for v_s , given that $V_{cc} = 15\text{ V}$ and $V_0 = 1\text{ V}$.

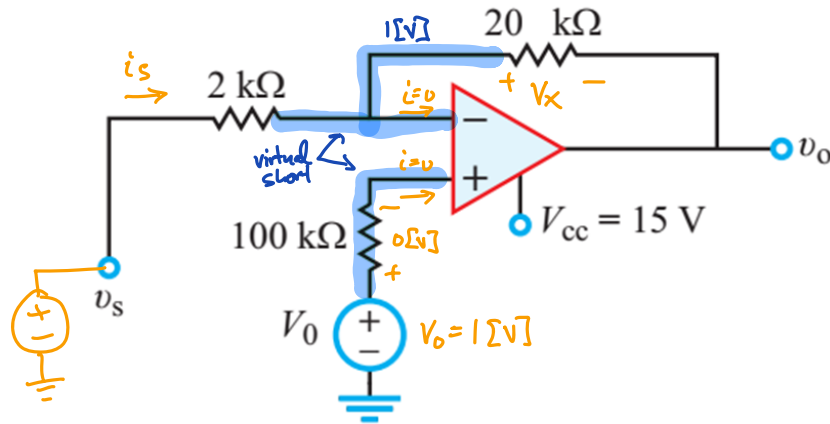


Figure P4.11 Circuit for Problems 4.11 and 4.12.

$$i_s = \frac{v_s - 1[\text{V}]}{2[\text{k}\Omega]}$$

$$V_x = i_s \cdot 20[\text{k}\Omega]$$

$$V_o = 1[\text{V}] - V_x = 1[\text{V}] - i_s \cdot 20[\text{k}\Omega]$$

$$= 1[\text{V}] - \frac{(v_s - 1[\text{V}])}{2[\text{k}\Omega]} \cdot 20[\text{k}\Omega]$$

$$= 1[\text{V}] - 10v_s + 10[\text{V}]$$

$$\boxed{V_o = -10v_s + 11[\text{V}]}$$

$$-15[\text{V}] < V_o < +15[\text{V}]$$

$$-15[\text{V}] < -10v_s + 11[\text{V}] < +15[\text{V}]$$

$$-26[\text{V}] < -10v_s < +4[\text{V}]$$

$$\boxed{-0.4[\text{V}] < v_s < +2.6[\text{V}]}$$